

Wellbore Model

Pressure Loss Equation

The equation governing the change in pressure in the direction of flow used for the wellbore model is

$$\Delta p = \Delta p_H - \Delta p_{KE} - \Delta p_F \quad (A3.1)$$

where

$$\Delta p_H = \rho g \Delta z \quad = \text{ (Hydrostatic head over the length } \Delta z \text{)}$$

$$\Delta p_F = \frac{2 f v^2 \rho}{D} \Delta z \quad = \text{ (Friction gain over a length } D z \text{)}$$

$$\Delta p_{KE} = -\rho v^2 \ln \left(\frac{p_2}{p_1} \right) \quad = \text{ (Kinetic energy gain)}$$

Here

ρ = density of the flowing "in situ" mixture

g = acceleration due to gravity

f = fanning friction factor

v = average velocity of mixture

D = inside pipe diameter

Δp = $p_2 - p_1$ = pressure drop over the length Δz

The wellbore model is a modification of the method of [Aziz et al. \(1972\)](#). The modifications involve the use of an EOS to calculate the phase behavior and fluid properties of the flowing fluid. Further details may be found in [Agarwal and Li, \(1988\)](#).

